



BT 101: Fundamentals of Biotechnology

Course Code	BT 101
Course Name	Fundamentals of Biotechnology
L-T-P-C	2-0-0-2
Prerequisites	NIL
Approval	5th Senate (Oct 2013)

Course Contents

- **Basic Component:** Introduction, structural and chemical components of cell;
- **Molecular mechanism and engineering:** DNA replication, RNA and protein synthesis, mutation and genetic recombination, Protoplast and cell fusion technology, Genetic engineering and PCR;
- **Microbes & fermentation:** Bacteria, fungi, viruses, microbial growth and bioprocess technology, Enzyme kinetics and bioconversion, Immobilization of enzymes, cell and application;
- **Plant and animal biotechnology and safety issues:** Biodiversity, transgenic plants and animals, Biosafety and bioethics;
- **Applications:** Biofuels, Bioremediation, bio-medical applications.

References

1. Smith J. E., **Biotechnology**, 3rd Edition, Cambridge University Press.
2. Walker J. M. and Gingold E. B., **Molecular Biology and Biotechnology**, The Royal Society of Chemistry, UK/Panama Publishing Corporation.
3. Auderisk G. and Auderisk T., **Biology life on Earth**, Macmillan Publishing Company.